SEMINAR REPORT

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POST QUANTUM CRYPTOGRAPHY

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BONAFIDE CERTIFICATE

This is to certify that this is a Bonafide record of seminar work on **"POST QUANTUM CRYPTOGRAPHY"** carried out by **ANANTHARAMAN P.K – MGP-21NMC012** during the tenth semester towards the partial fulfillment of the requirements for the Degree of **Master of Computer Applications (Integrated)** of APJ Abdul Kalam Technological University, Thiruvananthapuram.

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DECLARATION

I, the undersigned, hereby declare that the seminar report **"POST QUANTUM CRYTOGRAPHY"** submitted for partial fulfillment of the requirements for the award of Degree of Master of Computer Applications(Integrated) of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by me under the supervision of **Er. Anoopa Ravindran**. This submission represents my ideas in my own words, and where ideas or words of others have been included I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact, or source in my submission.

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ABSTRACT

Post-Quantum Cryptography (PQC), also known as quantum-resistant or quantum safe cryptography, seeks to develop new cryptographic primitives that remain secure against both classical and quantum adversaries. These schemes are designed to operate on existing classical hardware while ensuring resilience to quantum attacks. Post-Quantum Cryptography (PQC) aims to develop new cryptographic schemes resistant to quantum attacks while maintaining efficiency on classical hardware. The rapid advancements in quantum computing are expected to fundamentally disrupt conventional cryptographic systems that underpin global information security. Classical public-key algorithms such as RSA and Elliptic Curve Cryptography (ECC) derive their strength from the computational difficulty of problems like integer factorization and discrete logarithms. However, the emergence of quantum algorithms poses an existential threat to these schemes by enabling their efficient solution on sufficiently powerful quantum processors. As quantum computing technology progresses, the transition to PQC becomes imperative for ensuring the long-term security and reliability of digital systems. Post-Quantum Cryptography represents a proactive effort to safeguard data confidentiality, authentication, and integrity in the post-quantum era, where traditional cryptographic assumptions are no longer sufficient.

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CHAPTER 1
INTRODUCTION

1.1 Introduction to Crytography

Cryptography is the science and practice of securing information by transforming it into a protected format that can only be accessed by authorized parties. It plays a crucial role in ensuring secure communication in the presence of adversaries who may attempt to intercept or alter data. Cryptography is built on mathematical algorithms and secret keys that control the encryption and decryption processes.

The primary objectives of cryptography are confidentiality (preventing unauthorized access), integrity (ensuring data is not altered), authentication (verifying the identity of users), and non-repudiation (preventing denial of actions). Over time, cryptography has evolved from simple substitution ciphers to advanced public-key systems such as RSA and Elliptic Curve Cryptography (ECC), which are widely used in modern digital systems.

Cryptography forms the backbone of secure technologies including online banking, e-commerce, secure messaging, cloud computing, and digital signatures. With the rapid advancement of quantum computing, traditional cryptographic techniques face new security challenges, making the study of advanced and quantum-resistant cryptographic methods increasingly important.

1.2 Limitations of Classical Cryptography

Classical cryptographic algorithms such as RSA, Diffie–Hellman, and Elliptic Curve Cryptography rely on mathematical problems like integer factorization and discrete logarithms for their security. These problems are computationally hard for classical computers, which has made such algorithms secure for decades. However, their security assumptions do not hold in the presence of quantum computing.

With the development of quantum algorithms such as Shor's algorithm, these mathematical problems can be solved efficiently, making classical public-key cryptosystems vulnerable. Additionally, classical cryptography requires increasingly larger key sizes to maintain security, which leads to higher computational and storage overhead. As a result, classical cryptographic systems are not future-proof and may fail to protect long-term sensitive data against emerging quantum threats.

1.3 Quantum Computing Threats

Quantum computing poses a significant threat to current cryptographic systems due to its ability to solve certain mathematical problems much faster than classical com-

puters. Quantum algorithms such as Shor's algorithm can efficiently break widely used public-key cryptographic schemes like RSA, Diffie–Hellman, and Elliptic Curve Cryptography. Similarly, Grover's algorithm can reduce the effective security of symmetric encryption by speeding up brute-force attacks.

As quantum computing technology advances, encrypted data protected by classical algorithms may become vulnerable, leading to potential breaches of sensitive information. This threat makes existing cryptographic systems unsuitable for long-term security and highlights the urgent need for quantum-resistant cryptographic solutions such as Post-Quantum Cryptography.

1.4 Introduction to Post-Quantum Cryptography

Post-Quantum Cryptography (PQC) is a field of cryptography that focuses on designing secure cryptographic algorithms capable of withstanding attacks from quantum computers. Current public-key cryptographic systems such as RSA, Diffie–Hellman, and Elliptic Curve Cryptography depend on mathematical problems that are difficult for classical computers but can be efficiently solved using quantum algorithms like Shor's algorithm. This creates a serious security risk in the presence of large-scale quantum computers.

To address this challenge, PQC introduces new cryptographic techniques based on mathematical problems that are believed to be resistant to both classical and quantum attacks. These include lattice-based, hash-based, code-based, and multivariate polynomial cryptography. Unlike quantum cryptography, Post-Quantum Cryptography does not require quantum hardware and can be implemented using existing classical systems. PQC is therefore essential for ensuring long-term security of digital communication, data storage, and critical information infrastructure in the future quantum era.

CHAPTER 2
GENESIS AND EVOLUTION OF PQC

2.1 Early Quantum Algorithms

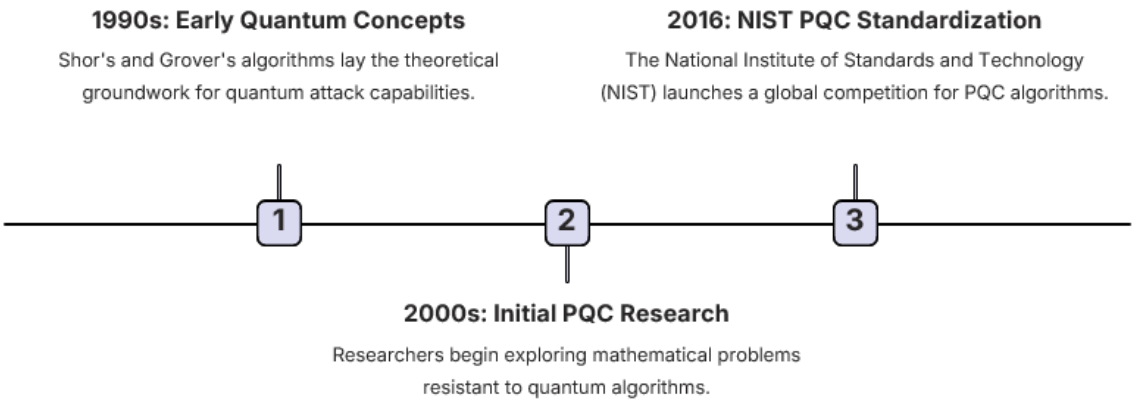
Early quantum algorithms played a foundational role in demonstrating the extraordinary computational capabilities of quantum computers and their profound implications for modern cryptography. These algorithms showed that certain mathematical problems, which are considered computationally infeasible for classical computers, can be solved efficiently using quantum principles such as superposition and entanglement. As a result, they highlighted serious vulnerabilities in widely used cryptographic systems and reshaped long-term security assumptions.

One of the most influential early quantum algorithms is Shor's algorithm, which can efficiently factor large integers and compute discrete logarithms in polynomial time. These problems form the security basis of classical public-key cryptographic systems such as RSA, Diffie–Hellman key exchange, and Elliptic Curve Cryptography (ECC). On classical computers, solving these problems requires impractical amounts of time for sufficiently large key sizes, ensuring security. However, Shor's algorithm breaks this assumption by enabling rapid solutions once a sufficiently powerful quantum computer becomes available. This poses a direct and serious threat to public-key infrastructures that secure digital communications, online transactions, and authentication systems worldwide.

Together, Shor's and Grover's algorithms exposed fundamental limitations of classical cryptographic techniques in the presence of quantum computing. They demonstrated that simply increasing key sizes is not always sufficient to guarantee security against quantum attacks, particularly for public-key systems. These revelations motivated extensive global research into cryptographic methods that remain secure in the quantum era. This research effort ultimately led to the development of Post-Quantum Cryptography, which focuses on designing cryptographic algorithms based on mathematical problems believed to be resistant to both classical and quantum attacks, ensuring long-term security for future digital systems.

In conclusion, early quantum algorithms fundamentally changed the understanding of computational security by revealing the weaknesses of classical cryptographic systems. The capabilities demonstrated by Shor's and Grover's algorithms made it clear that existing cryptographic methods cannot provide long-term security in the presence of quantum computers. These insights underscored the urgent need for quantum-resistant solutions and laid the groundwork for the development and adoption of Post-Quantum Cryptography as a critical component of future secure communication systems.

2.2 Evolution of Post-Quantum Cryptography



The evolution of Post-Quantum Cryptography began in the 1990s with the development of early quantum algorithms such as Shor's and Grover's algorithms, which demonstrated that quantum computers could break many classical cryptographic systems. These discoveries laid the theoretical foundation for quantum attack capabilities.

During the 2000s, researchers started exploring new mathematical problems that could remain secure against quantum attacks. This phase marked the beginning of initial Post-Quantum Cryptography research, focusing on lattice-based, code-based, and hash-based cryptographic approaches.

In 2016, the field gained global recognition when the National Institute of Standards and Technology (NIST) launched its Post-Quantum Cryptography standardization process. This initiative aimed to evaluate, select, and standardize quantum-resistant cryptographic algorithms for practical deployment, marking a major milestone in the evolution of PQC.

CHAPTER 3
ARCHITECTURE AND WORKING

3.1 System Architecture of Post-Quantum Cryptography

The system architecture of Post-Quantum Cryptography (PQC) is designed to provide secure communication in the presence of both classical and quantum computing threats. It follows a structure similar to traditional public-key cryptographic systems but replaces vulnerable algorithms with quantum-resistant ones. The core components of the architecture include key generation, encryption or digital signature generation, secure data transmission, and decryption or signature verification.

In a PQC system, cryptographic keys are generated using hard mathematical problems such as lattice structures, error-correcting codes, or hash-based constructions. The public key is shared over an insecure channel, while the private key is kept secret by the user. The sender uses the receiver's public key to encrypt data or verify authenticity, and the receiver uses the private key to decrypt the data or validate the signature.

These cryptographic operations are integrated into existing communication protocols, such as secure web communication and data storage systems, ensuring backward compatibility with current infrastructure. The architecture ensures essential security properties such as confidentiality, integrity, authentication, and non-repudiation, while also providing long-term protection against future quantum attacks.

Components of Post-Quantum Cryptography System Architecture:

1. **Key Generation**

Key generation is the first and most critical component of a PQC system. In this stage, a pair of cryptographic keys—public key and private key—is generated using quantum-resistant mathematical problems such as lattice-based constructions, error-correcting codes, or hash-based functions. The public key can be safely shared, while the private key is securely stored by the user. These keys form the foundation of secure communication in PQC.

2. **Encryption / Digital Signature Generation**

In this component, the sender uses the receiver's public key to encrypt the message or generate a digital signature. Encryption ensures that only the intended receiver can access the data, while digital signatures provide authenticity and integrity. PQC encryption and signature schemes are designed to remain secure even against quantum computer attacks.

3. **Secure Data Transmission**

Once encrypted or signed, the data is transmitted over an insecure communication channel such as the internet. Even if an attacker intercepts the transmitted data, they cannot extract useful information without the private key. PQC protects data during transmission from both classical and future quantum-based attacks.

4. **Decryption / Signature Verification**

At the receiver's end, the encrypted data is decrypted using the private key, or the digital signature is verified using the sender's public key. This step ensures that the data is readable only by authorized users and confirms that the message has not been altered during transmission.

5. **Integration with Existing Systems**

PQC components are designed to integrate smoothly with existing security protocols such as TLS, secure storage systems, and authentication frameworks. This ensures backward compatibility with current infrastructure while upgrading security to be quantum-resistant.

3.2 Working Principles of PQC Algorithms

Step 1: **Identification of Security Requirements**

Before implementing PQC, the security requirements of the system are identified, such as the level of confidentiality, type of data, performance constraints, and expected lifetime of the protected information. Based on these requirements, an appropriate Post-Quantum Cryptographic algorithm is selected.

Step 2: **Selection of PQC Algorithm**

A suitable PQC algorithm is chosen from categories such as lattice-based, hash-based, code-based, or multivariate cryptography. These algorithms are selected because they are resistant to known quantum attacks and are recommended by standardization bodies like NIST.

Step 3: **Key Generation**

The receiver generates a public key and a private key using the selected PQC algorithm. These keys are produced using complex mathematical structures such as lattices or error-correcting codes, which are computationally infeasible to break even using quantum computers.

Step 4: **Public Key Distribution**

The public key is distributed to the sender over an insecure channel. Since the public key does not reveal any information about the private key, it can be safely shared without compromising security and the private key remains confidential and secured.

Step 5: **Encryption or Digital Signature Verification**

Using the receiver's public key, the sender encrypts the message or verifies a digital signature. Encryption ensures that only the intended receiver can read the message, while digital signatures provide authentication and message integrity.

Step 6: **Secure Data Transmission**

The encrypted message or signed data is transmitted through communication networks such as the internet. Even if attackers intercept the transmitted data, they cannot extract useful information without the corresponding private key.

Step 7: **Decryption or Signature Generation**

Upon receiving the encrypted data, the receiver uses the private key to decrypt the message or generate a digital signature for verification. This step confirms that the message originates from a legitimate sender and has not been modified during transmission.

Step 8: **Integrity and Authenticity Verification**

The decrypted data is checked to ensure that it has not been altered and that it originates from a trusted source. This step guarantees data integrity and authentication.

Step 9: **Secure Storage and Key Management**

Keys and decrypted data are securely stored using proper key management techniques. Regular key updates and secure storage mechanisms are used to maintain long-term security.

Step 10: **Long-Term Quantum Security Assurance**

By using quantum-resistant mathematical foundations, PQC ensures long-term security against both current and future quantum computing threats, protecting sensitive data for extended periods.

CHAPTER 4
PQC ALGORITHM CATEGORIES

4.1 Lattice-Based Cryptography

Lattice-based cryptography is one of the most important approaches in Post-Quantum Cryptography. It is based on mathematical structures called lattices, which are regular grids of points in multi-dimensional space. The security of these schemes relies on hard lattice problems such as the Shortest Vector Problem (SVP) and Learning With Errors (LWE), which are believed to be difficult for both classical and quantum computers.

In lattice-based cryptographic systems, encryption and digital signatures are created by adding small random errors to mathematical equations. These errors make it extremely hard for an attacker to recover the secret key, even with advanced quantum algorithms. At the same time, authorized users can efficiently decrypt messages or verify signatures using their private keys.

Lattice-based cryptography offers strong security, fast performance, and flexibility, making it suitable for secure communication, digital signatures, and key exchange. Due to these advantages, many of the algorithms selected by NIST for Post-Quantum Cryptography standardization, such as CRYSTALS-Kyber and CRYSTALS-Dilithium, are based on lattice-based techniques.

4.2 Hash-Based Cryptography

Hash-based cryptography is a Post-Quantum Cryptography technique that secures data using cryptographic hash functions rather than complex mathematical problems. Its security depends on properties such as pre-image resistance, second pre-image resistance, and collision resistance, which are believed to remain strong even in the presence of quantum computers.

This cryptographic approach is primarily used for digital signature schemes, where hash functions help generate and verify signatures efficiently. Popular schemes such as Merkle Signature Schemes, XMSS, and SPHINCS+ organize hash values in tree structures to support secure and scalable signature generation.

Although hash-based cryptography often results in larger signature sizes and limited key usage compared to traditional schemes, it offers strong theoretical security and simplicity. Due to its proven reliability and resistance to quantum attacks, it is widely trusted for applications requiring long-term security, including secure authentication, software distribution, and data integrity protection.

4.3 Code-Based Cryptography

Code-based cryptography is a Post-Quantum Cryptography approach that is built on the theory of error-correcting codes. Its security relies on the difficulty of decoding a general linear code, known as the Syndrome Decoding Problem, which is considered hard for both classical and quantum computers. This makes code-based cryptography resistant to quantum attacks.

The most well-known code-based cryptosystem is the McEliece cryptosystem, proposed in 1978. It uses a public key generated from a large error-correcting code, typically a Goppa code, while the private key consists of the structure needed to decode the code efficiently. The public key appears random, making it extremely difficult for an attacker to determine the private key.

In the encryption process, the sender encodes the message using the receiver's public key and intentionally introduces a small number of errors. These errors protect the message from being decoded by unauthorized parties. The encrypted message is then transmitted over an insecure channel.At the receiver's end, the private key is used to remove the errors and decode the original message. At the receiver's end, the private key is used to remove the errors and decode the original message.

4.4 Multivariate Polynomial Cryptography

Multivariate Polynomial Cryptography is a Post-Quantum Cryptography approach that is based on the difficulty of solving systems of multivariate quadratic polynomial equations over finite fields. These problems, known as MQ problems, are considered computationally hard for both classical and quantum computers, making them suitable for quantum-resistant cryptographic systems.

In this approach, the private key consists of a set of structured polynomial equations that are easy to solve, while the public key is a transformed version of these equations that appears random and complex. The transformation hides the structure of the original system, preventing attackers from efficiently solving the equations to recover the private key.

Multivariate cryptography is mainly used for digital signature schemes due to its fast signing and verification processes.During signature generation, the signer uses the private key to solve the polynomial equations efficiently and produce a valid signature.

CHAPTER 5
SYSTEM DYNAMICS AND METHODOLOGY

5.1 Assessment of Existing Cryptographic Systems

Existing cryptographic systems are primarily designed to protect data using mathematical problems that are computationally difficult for classical computers to solve. Commonly used public-key cryptographic algorithms include RSA, Diffie–Hellman, and Elliptic Curve Cryptography (ECC), which rely on integer factorization and discrete logarithm problems. Symmetric algorithms such as AES and hash functions like SHA-2 and SHA-3 are also widely used for encryption, data integrity, and authentication.

While these cryptographic systems have proven effective against classical attacks, their security assumptions change significantly in the presence of quantum computing. Algorithms such as Shor's algorithm can efficiently break RSA, Diffie–Hellman, and ECC by solving factorization and discrete logarithm problems in polynomial time. This makes current public-key systems vulnerable to future quantum attacks. Symmetric algorithms and hash functions are less affected, but Grover's algorithm reduces their effective security strength, requiring larger key sizes to maintain security.

Another important concern in existing cryptographic systems is long-term data security. Data encrypted today using classical algorithms may be recorded by attackers and decrypted later once quantum computers become practical, a threat known as "harvest now, decrypt later." Additionally, key management, scalability, and compatibility issues arise when upgrading cryptographic systems to withstand quantum threats.

Performance and infrastructure limitations also affect current cryptographic systems. While classical algorithms are efficient and well-optimized, transitioning them to quantum-resistant versions requires changes in protocols, increased key sizes, and additional computational resources. Therefore, assessing existing cryptographic systems highlights the urgent need to adopt Post-Quantum Cryptography to ensure long-term security, reliability, and resilience against emerging quantum computing capabilities.

5.2 Algorithm Selection Criteria

The selection of cryptographic algorithms is a critical step in designing secure and efficient Post-Quantum Cryptography systems. Since PQC algorithms vary in structure, performance, and security assumptions, careful evaluation is required to ensure long-term protection against both classical and quantum attacks. The chosen algorithm must provide strong security guarantees while remaining practical for real-world implementation.

Algorithm Selection:

Step 1: **Evaluate Security Strength**

The first step in selecting a PQC algorithm is to evaluate its security strength. The algorithm must be resistant to both classical and quantum attacks. It should be based on strong mathematical problems with no known efficient attack methods. Security levels recommended by standardization bodies such as NIST are considered to ensure long-term protection.

Step 2: **Analyze Performance Metrics**

Next, the performance of the algorithm is analyzed. This includes the time taken for key generation, encryption, decryption, signing, and verification. The size of keys and signatures is also evaluated, as larger sizes may increase storage and communication overhead. Performance is especially important for systems with limited computational resources.

Step 3: **Check Integration Complexity**

The third step is to assess how easily the algorithm can be integrated into existing systems. The algorithm should work smoothly with current cryptographic protocols and software without requiring major changes. Easy integration reduces deployment cost and minimizes the risk of implementation errors.

Step 4: **Review Standardization and Maturity**

Finally, the algorithm's standardization status and maturity are reviewed. Algorithms that are part of recognized standardization efforts, such as NIST's Post-Quantum Cryptography project, are preferred. These algorithms have undergone extensive testing and peer review, making them more reliable and widely accepted.

5.3 Testing and Validation Procedures

Testing and validation are essential steps in ensuring that Post-Quantum Cryptography (PQC) algorithms are secure, reliable, and practical for real-world deployment. These algorithms must be rigorously evaluated to resist both classical and quantum attacks. The evaluation process includes verifying correctness, assessing performance efficiency, and identifying potential vulnerabilities. Thorough testing not only ensures compliance with cryptographic standards but also builds confidence in their adoption across diverse applications, from secure communications to data protection in critical systems.

Functional Testing

The first stage involves functional testing to verify that the cryptographic algorithm works as intended. This includes testing key generation, encryption, decryption, digital signature creation, and signature verification. The system is checked to ensure that valid inputs produce correct outputs and that errors are handled securely without leaking sensitive information.

Security Testing

Security testing evaluates the resistance of PQC algorithms to known attack models. This includes testing against classical cryptographic attacks such as brute force, side-channel attacks, and fault injection attacks. Additionally, the algorithm is assessed for its resistance to quantum-based attacks by analyzing its underlying mathematical hardness assumptions.

Performance Testing

Performance testing measures the efficiency of the PQC algorithm in practical environments. Metrics such as computation time, memory usage, key size, and communication overhead are evaluated. These tests help determine whether the algorithm is suitable for deployment in systems with limited resources, such as mobile devices, embedded systems, and IoT applications.

Compatibility and Integration Testing

This phase checks how well the PQC algorithm integrates with existing cryptographic protocols and infrastructure. The algorithm is tested within real-world systems such as secure communication protocols, databases, and authentication frameworks to ensure seamless interoperability and backward compatibility.

Stress and Scalability Testing

Stress testing evaluates the algorithm's behavior under heavy workloads and high traffic conditions. Scalability testing ensures that the algorithm performs reliably as the number of users, devices, or transactions increases. This helps identify potential performance bottlenecks or stability issues.

Compliance and Standard Validation

Finally, PQC algorithms are validated against international standards and guidelines, such as those set by NIST. Compliance testing ensures that the implementation follows recommended security practices and meets required security levels. Algorithms that pass these validation procedures are considered suitable for long-term deployment.

5.4 Performance Benchmarking

Performance benchmarking is a critical step in evaluating Post-Quantum Cryptography (PQC) algorithms to ensure they are practical for real-world deployment. Since PQC algorithms are generally more complex than classical cryptographic schemes, benchmarking helps assess their efficiency in terms of computation time, memory usage, and key sizes. It involves measuring how quickly algorithms can perform encryption, decryption, key generation, and digital signature operations across different hardware platforms, including servers, mobile devices, and resource-constrained Internet-of-Things (IoT) devices.

Benchmarking also evaluates the trade-offs between security and performance. For example, while some lattice-based algorithms provide strong quantum resistance, they may require larger keys or more processing power, which can affect system responsiveness. Hash-based and code-based algorithms may have different performance profiles in terms of signature size or verification speed. By comparing multiple algorithms under standardized conditions, researchers and organizations can select the most suitable PQC schemes for their specific applications.

Additionally, performance benchmarking helps identify potential bottlenecks, optimize implementations, and guide protocol design. It ensures that PQC algorithms meet the practical requirements of modern digital systems, such as low latency, high throughput, and energy efficiency, without compromising security. Regular benchmarking, combined with rigorous testing, provides confidence that chosen PQC solutions can withstand quantum threats while remaining efficient and deployable across diverse real-world scenarios.

Another important aspect of performance benchmarking is the evaluation of algorithm scalability and adaptability across different environments. PQC algorithms must perform efficiently not only on high-performance servers but also on constrained devices like smart cards, IoT sensors, and mobile platforms. Benchmarking under varied network conditions and real-world workloads helps determine the practical limits of each algorithm, including latency, bandwidth requirements, and energy consumption

CHAPTER 5
ADVANTAGES AND LIMITATIONS

6.1 Advantages of Post-Quantum Cryptography

Quantum-Resistant Security: PQC algorithms are designed to resist attacks from both classical and quantum computers, ensuring long-term protection of sensitive data.

Future-Proof Protection: PQC safeguards information against potential future quantum attacks, mitigating the "harvest now, decrypt later" threat.

Diverse Algorithmic Approaches: PQC employs multiple mathematical foundations—lattice-based, hash-based, code-based, and multivariate polynomial cryptography—providing flexibility and robustness.

Compatibility with Existing Infrastructure: Many PQC schemes can be integrated into current systems without major hardware changes, facilitating smoother adoption.

Efficient Performance: PQC algorithms can offer reasonable computational efficiency and key sizes suitable for real-world applications, including IoT, mobile devices, and cloud services.

6.2 Limitations and Challenges

Larger Key Sizes and Ciphertext: Many PQC algorithms, especially lattice-based and code-based schemes, require significantly larger keys and ciphertexts compared to classical cryptography, which can increase storage and transmission overhead.

Higher Computational Complexity: Some PQC algorithms demand more processing power, which can affect performance on resource-constrained devices like IoT sensors, embedded systems, and mobile devices.

Implementation Challenges: Implementing PQC securely requires careful attention to avoid side-channel attacks, coding errors, or improper integration into existing protocols.

Limited Standardization: Although the NIST PQC project is progressing, many algorithms are still under evaluation, and global standards are not fully finalized, which can create uncertainty for adoption.

Interoperability Issues: Transitioning from classical to post-quantum systems may lead to compatibility problems with legacy systems and protocols.

CHAPTER 7
APPLICATIONS AND FUTURE SCOPE

7.1 Applications of Post-Quantum Cryptography

Post-Quantum Cryptography (PQC) is designed to secure digital communication and data against the emerging threats posed by quantum computing. Its applications span multiple domains, addressing the need for future-proof security in environments where sensitive information must remain confidential, authenticated, and tamper-proof. By integrating PQC algorithms, organizations can ensure that their cryptographic systems remain resilient against both classical and quantum attacks.

Key applications include:

Secure Communications: PQC can be used to protect emails, messaging platforms, and voice/video calls from interception and decryption by quantum-enabled adversaries.

Digital Signatures and Authentication: PQC algorithms can generate quantum-resistant digital signatures to verify identity and ensure data integrity in transactions, contracts, and software updates.

Data Encryption for Storage and Cloud Services: Organizations can secure sensitive data stored in databases, cloud services, and distributed storage systems against future quantum threats.

Financial Transactions and Banking Security: PQC protects online banking, payment systems, and blockchain networks, ensuring secure transactions and preventing potential quantum attacks on financial systems.

Critical Infrastructure Protection: PQC can safeguard power grids, water supply systems, transportation networks, and other critical infrastructures against cyberattacks enabled by quantum computing.

IoT and Embedded Systems Security: PQC provides lightweight yet secure cryptographic solutions for IoT devices, smart sensors, and industrial control systems vulnerable to quantum threats.

Government and Defense Applications: PQC secures classified communications, defense networks, and sensitive government data, ensuring national security in a post-quantum era.

Long-Term Data Protection: PQC is critical for safeguarding information that must remain confidential for decades, such as medical records, legal documents, and intellectual property.

7.2 Future Scope

The future scope of Post-Quantum Cryptography (PQC) is closely tied to the rapid advancement of quantum computing, which threatens to compromise classical cryptographic systems. As quantum computers become more capable, adopting PQC will be essential across industries that handle sensitive data, including finance, healthcare, government, and critical infrastructure. This ensures that digital communications, transactions, and stored information remain secure against both current and future threats.

Standardization efforts, led by organizations like NIST, will pave the way for seamless integration of PQC into existing cryptographic protocols. The development of optimized algorithms will focus on improving computational efficiency, reducing key sizes, and enabling secure implementation on resource-constrained devices such as IoT devices and mobile platforms. Hybrid cryptography solutions, combining classical and quantum-resistant algorithms, are expected to be widely adopted during the transition phase to maintain backward compatibility.

Emerging technologies, such as blockchain, autonomous vehicles, smart cities, and AI systems, will increasingly rely on PQC for secure operations. Ongoing research into novel quantum-resistant algorithms will expand the capabilities of PQC, making it more versatile and robust. Additionally, global collaboration among governments, industry, and academia will establish best practices, policies, and regulations to guide secure deployment of PQC worldwide.

PQC also has significant implications for cloud computing and data storage. As organizations continue to store vast amounts of sensitive data in distributed cloud environments, integrating PQC ensures that this information remains secure even if intercepted or accessed by a quantum-enabled adversary. Similarly, PQC will play a key role in protecting critical infrastructure from future cyberattacks, ensuring the continuity of essential services such as power grids, water systems, and transportation networks.

CHAPTER 8
CONCLUSION

8.1 Conclusion

Post-Quantum Cryptography (PQC) is an essential advancement in the field of cybersecurity, designed to protect data against the growing threat of quantum computers. Traditional cryptographic systems, such as RSA and ECC, are vulnerable to attacks by powerful quantum machines. PQC provides a future-proof solution by using mathematical problems that are resistant to both classical and quantum attacks, ensuring the long-term security of sensitive information.

The development and implementation of PQC involve rigorous testing, standardization, and evaluation to ensure reliability, efficiency, and practicality. Multiple algorithms, including lattice-based, hash-based, code-based, and multivariate schemes, offer a range of options for securing digital communications, financial transactions, government data, and critical infrastructure. Adopting these algorithms helps organizations stay ahead of emerging cyber threats and maintain trust in their digital systems.

While PQC has many advantages, there are also challenges, such as larger key sizes, higher computational demands, and integration with existing systems. Research and optimization efforts are ongoing to overcome these limitations, making PQC more practical for real-world applications. The gradual adoption of hybrid solutions, combining classical and quantum-resistant algorithms, will help organizations transition smoothly to quantum-safe systems.

Looking ahead, PQC will play a crucial role in securing the digital world, from cloud computing and IoT devices to smart cities and blockchain technologies. By proactively implementing PQC, organizations can ensure the long-term protection of sensitive data and maintain resilience against future quantum attacks. Continuous research, global collaboration, and widespread awareness will further strengthen the adoption and effectiveness of Post-Quantum Cryptography in the years to come.

CHAPTER 9
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CHAPTER 10
APPENDIX

Post-Quantum Cryptography: Securing Data Beyond Classical Encryption

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Seminar Guide: Anoopa Ravindran

Abstract

Post-Quantum Cryptography (PQC), also known as quantum-resistant or quantum safe cryptography, seeks to develop new cryptographic primitives that remain secure against both classical and quantum adversaries. These schemes are designed to operate on existing classical hardware while ensuring resilience to quantum attacks. Post-Quantum Cryptography (PQC) aims to develop new cryptographic schemes resistant to quantum attacks while maintaining efficiency on classical hardware. The rapid advancements in quantum computing are expected to fundamentally disrupt conventional cryptographic systems that underpin global information security. Classical public-key algorithms such as RSA and Elliptic Curve Cryptography (ECC) derive their strength from the computational difficulty of problems like integer factorization and discrete logarithms. However, the emergence of quantum algorithms poses an existential threat to these schemes by enabling their efficient solution on sufficiently powerful quantum processors. As quantum computing technology progresses, the transition to PQC becomes imperative for ensuring the long-term security and reliability of digital systems. Post-Quantum Cryptography represents a proactive effort to safeguard data confidentiality, authentication, and integrity in the post-quantum era, where traditional cryptographic assumptions are no longer sufficient.

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Introduction to Post-Quantum Cryptography

Current public-key cryptography relies on the computational difficulty of factoring large numbers or solving discrete logarithms. Shor's algorithm, a quantum algorithm, can efficiently break these systems, rendering them obsolete.

This impending threat necessitates the development of new cryptographic primitives resistant to quantum attacks to protect sensitive data long-term.

As quantum computing technology progresses, the transition to PQC becomes imperative for ensuring the long-term security and reliability of digital systems. Post-Quantum Cryptography represents a proactive effort to safeguard data confidentiality, authentication, and integrity in the post-quantum era, where traditional cryptographic assumptions are no longer sufficient.

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Literature Survey: Foundations of PQC

1990s: Early Quantum Concepts

Shor's and Grover's algorithms lay the theoretical groundwork for quantum attack capabilities.

2016: NIST PQC Standardization

The National Institute of Standards and Technology (NIST) launches a global competition for PQC algorithms.

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2000s: Initial PQC Research

Researchers begin exploring mathematical problems resistant to quantum algorithms.

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PQC Algorithm Categories

Post-Quantum Cryptography focuses on mathematical problems that are conjectured to be intractable for both classical and quantum computers.

Lattice-based Cryptography

Based on hard problems in lattices, offering strong theoretical security guarantees.

Code-based Cryptography

Leverages error-correcting codes, like McEliece cryptosystem, for secure communication.

Hash-based Signatures

Provides robust digital signatures, often with larger key sizes but proven security.

Multivariate Polynomial

Utilizes systems of multivariate polynomials over finite fields, primarily for signatures.

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Lattice-based Cryptography

Lattice-based cryptography keeps information secure by using very difficult math problems. These problems are based on complex structures called "lattices." They are so hard that even supercomputers and future quantum computers cannot solve them easily. This makes lattice-based cryptography a strong defense against new quantum attacks.

Uses Hard Math Problems

Its security relies on math problems that are extremely difficult to solve, like finding specific points within complex 'lattices'.

Safe from Quantum Attacks

These math problems are too hard for both today's computers and future quantum computers, making it a powerful way to protect against new cyber threats.

Strong Security Tools

It provides solid ways to encrypt data (hide information) and create digital signatures (verify who sent something). This is key for future online safety.

Becoming a Standard

Key examples, like Kyber for encryption and Dilithium for signatures, are being reviewed by NIST. This means they are likely to become official standards for cybersecurity.

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Hash-based Cryptography

Hash-based cryptography uses special 'one-way' functions called hash functions. These functions are easy to create but extremely difficult to reverse. Unlike older digital security that relies on complex math problems, hash functions are generally thought to be safe from quantum computers.

One-Way Functions

Its security comes from how hard it is to create a fake message that looks real (collision resistance) or to figure out the original message from its hash (pre-image resistance). These are strong even against quantum attacks.

Quantum Resistant

These methods are proven to be safe from quantum attacks. They offer a strong promise to protect against future quantum hackers, without relying on new, untested ideas.

Digital Signatures

They are mainly used for digital signatures. These signatures prove that a message is real and hasn't been changed. Examples like XMSS and SPHINCS+ have been approved by NIST.

Practical Considerations

These methods often need larger digital 'keys' and signatures compared to older ways. Some versions also need careful management to stay secure.

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Code-based Cryptography

Code-based encryption gets its strength from how incredibly difficult it is to break special types of codes. These systems use error-correcting codes, which were originally made to fix mistakes when data is sent, to build strong and secure ways to protect information.

Hard to Break

Its security comes from how incredibly difficult it is for any computer, even future quantum ones, to break these complex codes.

Uses Error-Fixing Codes

This method uses codes that were first invented to fix errors when sending data. Now, these same principles are used for security.

McEliece System

The McEliece system is a well-known example. It has stayed secure for a long time, ever since it was first created in 1978.

Quantum-Safe, But Big

It offers strong protection against future quantum attacks. However, this often means using larger keys and encrypted messages.

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Multivariate Polynomial Cryptography

Multivariate Polynomial Cryptography (MPC) uses tricky math problems involving many variables to keep information safe. This math problem, called the Multivariate Quadratic (MQ) problem, is very hard to solve for any computer, even future quantum ones. This makes MPC a good choice for security that can stand up to quantum attacks.

How It Works

MPC systems rely on how difficult it is to find answers to math problems with many equations and unknowns. These problems usually use small number systems.

Safe from Quantum Computers

The main math problem (MQ) is tough for today's computers and future quantum computers. This means it offers strong protection against new types of attacks.

Main Uses

While MPC can be used for both locking and unlocking data, it's mostly used for digital signatures. This is because it can create and check signatures very quickly.

Things to Know

These systems often need larger public keys. They also need careful design to avoid certain math attacks. Experts are still working to make them faster and even more secure.

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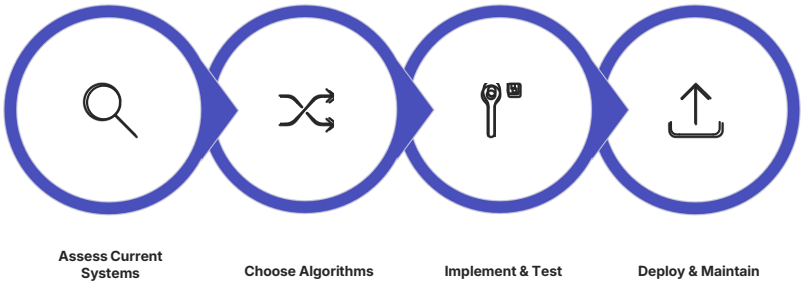
System Dynamics



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Assess Current Systems

Starting with Post-Quantum Cryptography (PQC) means first looking closely at current security setups. This important first step helps identify weak spots and determine the effort needed for a smooth change. It ensures a clear understanding of current security and shows the way forward to a quantum-safe future.

1. Review Security

The process begins with a careful check of all existing security systems to identify any weak points. This review helps to understand what needs to be done to switch to PQC.

2. List Crypto Tools

Next, a full list of all encryption tools and security programs is compiled. The interconnections and dependencies of these systems are also mapped out. This highlights the key areas that PQC integration will affect.

3. Categorize Data

Private information is categorized based on its secrecy level and required retention period for safety. This helps determine which data needs the strongest and fastest PQC protection, and for how long.

4. Check Rules and Laws

Finally, all important rules and laws are reviewed. This ensures that new security measures will meet all required standards. This way, operations remain safe from future threats and always follow the rules.

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Algorithm Selection: Choosing the Right PQC Schemes

Picking the best Post-Quantum Cryptography (PQC) algorithms is a key step. It's not just about how well they resist quantum attacks. Organizations need to check each possible algorithm. They must make sure it fits their security needs, how they operate, and their long-term goals.

Security Strength

Check how strong the algorithm is against both current (classical) and future (quantum) attacks. Consider the security levels set by NIST and any known weaknesses.

Performance Metrics

Look at how fast the algorithm works in real life. This includes how quickly it creates keys, encrypts or decrypts data, and signs or checks signatures. Also, consider the size of the keys and signatures, especially on different computer systems.

Integration Complexity

Think about how easy it will be to add new PQC algorithms to your current systems and tools. The goal is to cause as little trouble as possible and make sure everything still works together smoothly.

Standardization & Maturity

Choose algorithms that are part of international standardization programs, like NIST's PQC project. This means they have been thoroughly reviewed and are more likely to be stable and widely used for a long time.

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PQC Testing Procedures: Ensuring Effective Implementation

Using Post-Quantum Cryptography (PQC) involves more than just picking the right algorithms. These algorithms must also be carefully integrated into existing systems and thoroughly tested. This step ensures that the chosen PQC methods are not only safe from quantum attacks but also function well with current systems. The objective is to avoid slowing things down or creating new weak points.

Phased Rollout

PQC algorithms are initially introduced slowly, into separate systems or less important applications. This approach allows for careful testing and understanding of any risks before enterprise-wide deployment.

Compatibility Verification

PQC is thoroughly tested with all existing security tools, rules, and programs. The goal is to ensure seamless operation without causing disruptions to daily operations.

Performance Measurement

The extra computing power PQC requires, response times, and data handling capacity are all assessed. Settings are fine-tuned to ensure users do not experience any slowdowns and systems run efficiently.

Comprehensive Security Audits

Complete security reviews, penetration tests, and vulnerability assessments are conducted. This confirms that the PQC is robust and that its integration does not introduce new attack vectors.

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Deploy and Maintain

After new PQC methods are tested, they are implemented. Regular checks and updates are also maintained. This helps ensure systems remain safe from future quantum attacks and can be easily adapted when new dangers or solutions emerge.

Introduce it Slowly

PQC solutions are introduced step by step. Implementation begins with systems that are less critical or with small test groups. This helps avoid big problems and allows for changes to be made based on lessons learned before widespread deployment of PQC.

Keep a Close Eye on It

Robust monitoring systems are established to constantly oversee PQC operations. Metrics such as speed, computational resource utilization, and potential vulnerabilities are monitored. This helps identify problems early.

Manage Security Keys Carefully

Strict rules are established for the management of PQC 'keys.' These are like secret codes. Their creation, sharing, storage, and revocation processes are defined. The safeguarding of these keys is critical for PQC to function correctly.

Be Ready for Changes

Systems are designed so PQC methods can be easily updated or swapped out. This is important because new security rules might appear, or new weaknesses could be found. Regular updates help maintain safety from future quantum threats.

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Methodology: Performance Benchmarking

A method for testing performance helps in evaluating different Post-Quantum Cryptography (PQC) methods in a full and fair way, using real-world conditions.

1 **Metrics Measured**
Measurements include factors such as key generation time, data encryption/decryption speed, digital signature creation and verification speed, and the size of keys and protected data at various security levels.

2 **Consistent Testing Setup**
All tests are run on the same computer setup, using specific types of processors, operating systems, and software tools. This ensures that anyone can repeat the tests and get the same results.

3 **Careful Testing Steps**
Tests are repeated many times, warm-up runs are performed, and statistical analysis is used to get reliable data. This helps reduce outside influences and account for any system quirks.

4 **Comparing Results**
PQC methods are compared against each other and against older, well-known encryption types. This reveals the pros and cons in terms of speed, resource use, and security.

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Pros and Cons Of Post-Quantum Crypotography

Moving to quantum-safe encryption (PQC) helps protect our digital information for the future. But it also comes with new challenges we need to think about.

Pros

- **Quantum-Proof Security:** It keeps sensitive data safe from super-powerful quantum computers that might exist in the future.
- **Lasting Data Protection:** This new security helps keep your information private and correct for many years to come.
- **Easier to Adapt:** It makes our computer systems more flexible, so they can handle new security updates more easily.

Cons

- **Slightly Slower:** These new methods can sometimes use bigger keys, take longer to encrypt or decrypt data, and require more computer power.
- **Tricky to Implement:** It means we have to make big changes to our current computer systems and software.
- **Still Developing:** Some of the new encryption ideas are quite new and are still being researched and made official.

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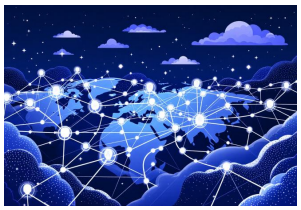
Applications and Future Directions

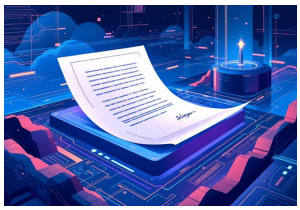
Secure Communications

Ensuring confidentiality and integrity for sensitive government, financial, and personal data.



Digital Signatures

Protecting software updates, blockchain transactions, and authenticated access from quantum adversaries.



Long-Term Data Archiving

Preserving the secrecy of encrypted data that must remain confidential for decades.

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Conclusion

Post-Quantum Cryptography is not just an academic exercise but a strategic imperative for global cybersecurity. The transition to quantum-resistant algorithms is a complex, multi-year endeavor requiring significant research, development, and international collaboration.

Proactive Defense

Essential for protecting against future quantum threats.

Ongoing Development

Algorithms are continually refined and standardized.

Global Adoption

Requires widespread implementation across industries.

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